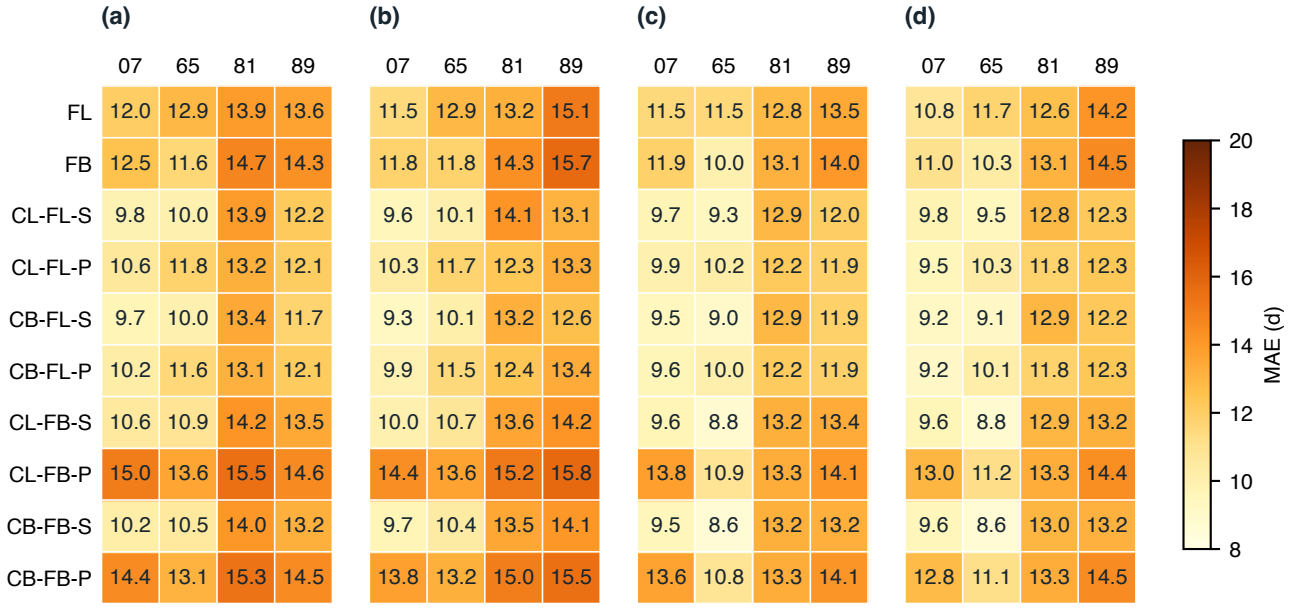
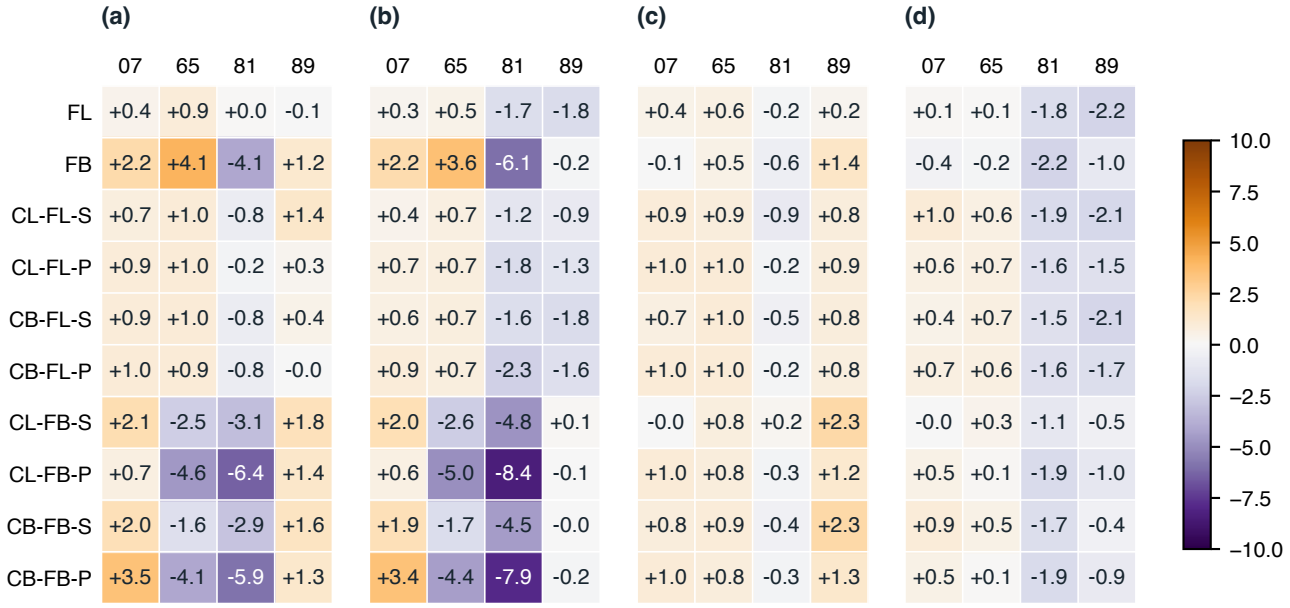
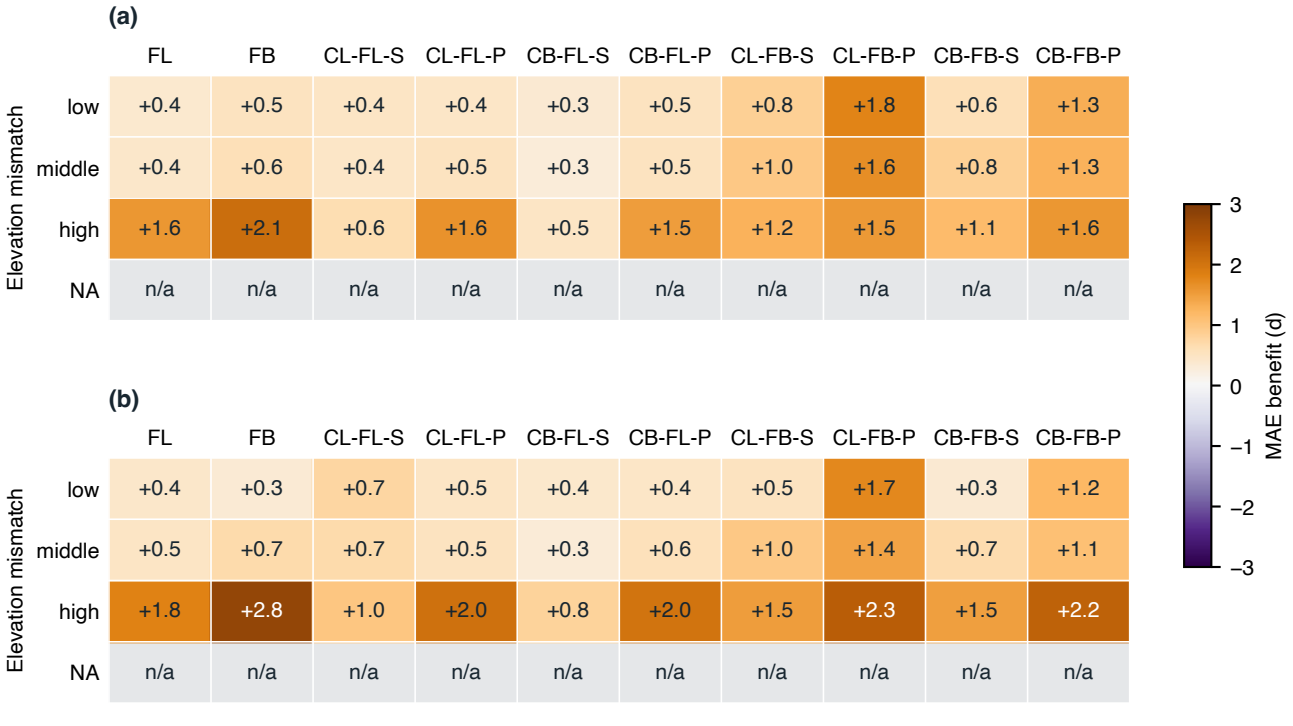
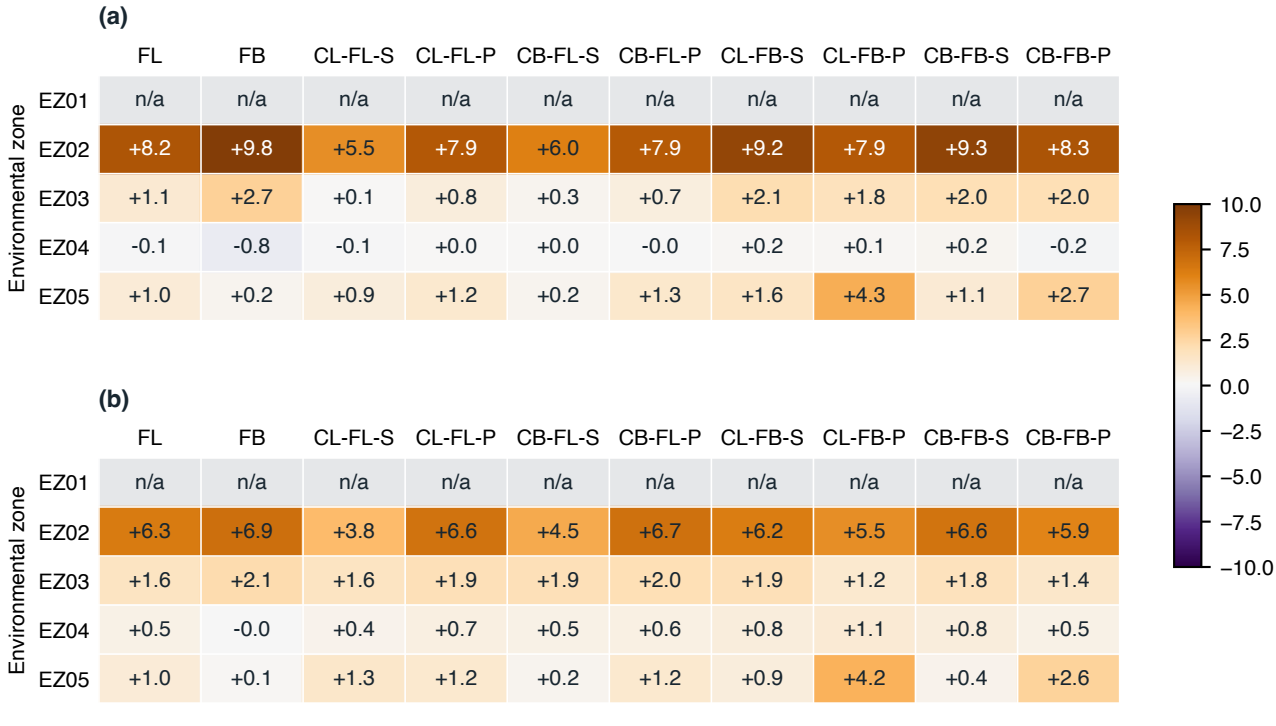

(a)
(b)
(c)
(d)
07 65 81 89
FL 12.0 12.9 13.9 13.6 11.5 12.9 13.2 15.1 11.5 11.5 12.8 13.5 10.8 11.7 12.6 14.2
FB 12.5 11.6 14.7 14.3 11.8 11.8 14.3 15.7 11.9 10.0 13.1 14.0 11.0 10.3 13.1 14.5
CL-FL-S 9.8 10.0 13.9 12.2 9.6 10.1 14.1 13.1 9.7 9.3 12.9 12.0 9.8 9.5 12.8 12.3
CL-FL-P 10.6 11.8 13.2 12.1 10.3 11.7 12.3 13.3 9.9 10.2 12.2 11.9 9.5 10.3 11.8 12.3
CB-FL-S 9.7 10.0 13.4 11.7 9.3 10.1 13.2 12.6 9.5 9.0 12.9 11.9 9.2 9.1 12.9 12.2
CB-FL-P 10.2 11.6 13.1 12.1 9.9 11.5 12.4 13.4 9.6 10.0 12.2 11.9 9.2 10.1 11.8 12.3
CL-FB-S 10.6 10.9 14.2 13.5 10.0 10.7 13.6 14.2 9.6 8.8 13.2 13.4 9.6 8.8 12.9 13.2
CL-FB-P 15.0 13.6 15.5 14.6 14.4 13.6 15.2 15.8 13.8 10.9 13.3 14.1 13.0 11.2 13.3 14.4
CB-FB-S 10.2 10.5 14.0 13.2 9.7 10.4 13.5 14.1 9.5 8.6 13.2 13.2 9.6 8.6 13.0 13.2
CB-FB-P 14.4 13.1 15.3 14.5 13.8 13.2 15.0 15.5 13.6 10.8 13.3 14.1 12.8 11.1 13.3 14.5
MAE (d)
20
18
16
14
12
10
8


(a)
(b)
(c)
(d)
07 65 81 89
FL +0.4 +0.9 +0.0 -0.1 +0.3 +0.5 -1.7 -1.8 +0.4 +0.6 -0.2 +0.2 +0.1 +0.1 -1.8 -2.2
FB +2.2 +4.1 -4.1 +1.2 +2.2 +3.6 -6.1 -0.2 -0.1 +0.5 -0.6 +1.4 -0.4 -0.2 -2.2 -1.0
CL-FL-S +0.7 +1.0 -0.8 +1.4 +0.4 +0.7 -1.2 -0.9 +0.9 +0.9 -0.9 +0.8 +1.0 +0.6 -1.9 -2.1
CL-FL-P +0.9 +1.0 -0.2 +0.3 +0.7 +0.7 -1.8 -1.3 +1.0 +1.0 -0.2 +0.9 +0.6 +0.7 -1.6 -1.5
CB-FL-S +0.9 +1.0 -0.8 +0.4 +0.6 +0.7 -1.6 -1.8 +0.7 +1.0 -0.5 +0.8 +0.4 +0.7 -1.5 -2.1
CB-FL-P +1.0 +0.9 -0.8 -0.0 +0.9 +0.7 -2.3 -1.6 +1.0 +1.0 -0.2 +0.8 +0.7 +0.6 -1.6 -1.7
CL-FB-S +2.1 -2.5 -3.1 +1.8 +2.0 -2.6 -4.8 +0.1 -0.0 +0.8 +0.2 +2.3 -0.0 +0.3 -1.1 -0.5
CL-FB-P +0.7 -4.6 -6.4 +1.4 +0.6 -5.0 -8.4 -0.1 +1.0 +0.8 -0.3 +1.2 +0.5 +0.1 -1.9 -1.0
CB-FB-S +2.0 -1.6 -2.9 +1.6 +1.9 -1.7 -4.5 -0.0 +0.8 +0.9 -0.4 +2.3 +0.9 +0.5 -1.7 -0.4
CB-FB-P +3.5 -4.1 -5.9 +1.3 +3.4 -4.4 -7.9 -0.2 +1.0 +0.8 -0.3 +1.3 +0.5 +0.1 -1.9 -0.9
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
Bias (d)


(a)
FL
FB
CL-FL-S
CL-FL-P
CB-FL-S
CB-FL-P
CL-FB-S
CL-FB-P
CB-FB-S
CB-FB-P
Terrain variability
low
middle
high
NA
+0.9
+1.1
+0.5
+1.0
+0.4
+1.0
+1.1
+2.0
+0.9
+1.6
+0.6
+0.9
+0.4
+0.7
+0.3
+0.7
+1.2
+1.8
+1.1
+1.5
+0.8
+1.1
+0.6
+0.8
+0.4
+0.8
+0.7
+1.1
+0.5
+0.9
n/a
(b)
+1.1
+1.4
+0.9
+1.1
+0.6
+1.1
+1.6
+2.3
+1.4
+1.9
+1.0
+1.2
+1.2
+1.2
+0.4
+1.1
+0.9
+2.1
+0.8
+1.7
+0.4
+0.9
+0.3
+0.6
+0.3
+0.6
+0.2
+0.9
+0.2
+0.7
MAE benefit (d)
3
2
1
0
−1
−2
−3


(a)
FL
FB
CL-FL-S
CL-FL-P
CB-FL-S
CB-FL-P
CL-FB-S
CL-FB-P
CB-FB-S
CB-FB-P
Elevation mismatch
low
middle
high
NA
+0.4 +0.5 +0.4 +0.4 +0.3 +0.5 +0.8 +1.8 +0.6 +1.3
+0.4 +0.6 +0.4 +0.5 +0.3 +0.5 +1.0 +1.6 +0.8 +1.3
+1.6 +2.1 +0.6 +1.6 +0.5 +1.5 +1.2 +1.5 +1.1 +1.6
n/a n/a n/a n/a n/a n/a n/a n/a n/a n/a
(b)
FL
FB
CL-FL-S
CL-FL-P
CB-FL-S
CB-FL-P
CL-FB-S
CL-FB-P
CB-FB-S
CB-FB-P
Elevation mismatch
low
middle
high
NA
+0.4 +0.3 +0.7 +0.5 +0.4 +0.4 +0.5 +1.7 +0.3 +1.2
+0.5 +0.7 +0.7 +0.5 +0.3 +0.6 +1.0 +1.4 +0.7 +1.1
+1.8 +2.8 +1.0 +2.0 +0.8 +2.0 +1.5 +2.3 +1.5 +2.2
n/a n/a n/a n/a n/a n/a n/a n/a n/a n/a
MAE benefit (d)
3
2
1
0
−1
−2
−3

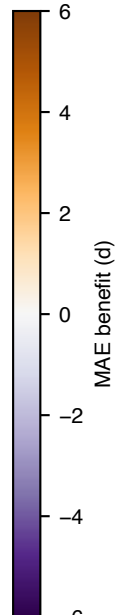

(a)
FL
FB
CL-FL-S
CL-FL-P
CB-FL-S
CB-FL-P
CL-FB-S
CL-FB-P
CB-FB-S
CB-FB-P
Environmental zone
EZ01 n/a n/a n/a n/a n/a n/a n/a n/a n/a n/a
EZ02 +8.2 +9.8 +5.5 +7.9 +6.0 +7.9 +9.2 +7.9 +9.3 +8.3
EZ03 +1.1 +2.7 +0.1 +0.8 +0.3 +0.7 +2.1 +1.8 +2.0 +2.0
EZ04 -0.1 -0.8 -0.1 +0.0 +0.0 -0.0 +0.2 +0.1 +0.2 -0.2
EZ05 +1.0 +0.2 +0.9 +1.2 +0.2 +1.3 +1.6 +4.3 +1.1 +2.7
(b)
FL
FB
CL-FL-S
CL-FL-P
CB-FL-S
CB-FL-P
CL-FB-S
CL-FB-P
CB-FB-S
CB-FB-P
Environmental zone
EZ01 n/a n/a n/a n/a n/a n/a n/a n/a n/a n/a
EZ02 +6.3 +6.9 +3.8 +6.6 +4.5 +6.7 +6.2 +5.5 +6.6 +5.9
EZ03 +1.6 +2.1 +1.6 +1.9 +1.9 +2.0 +1.9 +1.2 +1.8 +1.4
EZ04 +0.5 -0.0 +0.4 +0.7 +0.5 +0.6 +0.8 +1.1 +0.8 +0.5
EZ05 +1.0 +0.1 +1.3 +1.2 +0.2 +1.2 +0.9 +4.2 +0.4 +2.6
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0

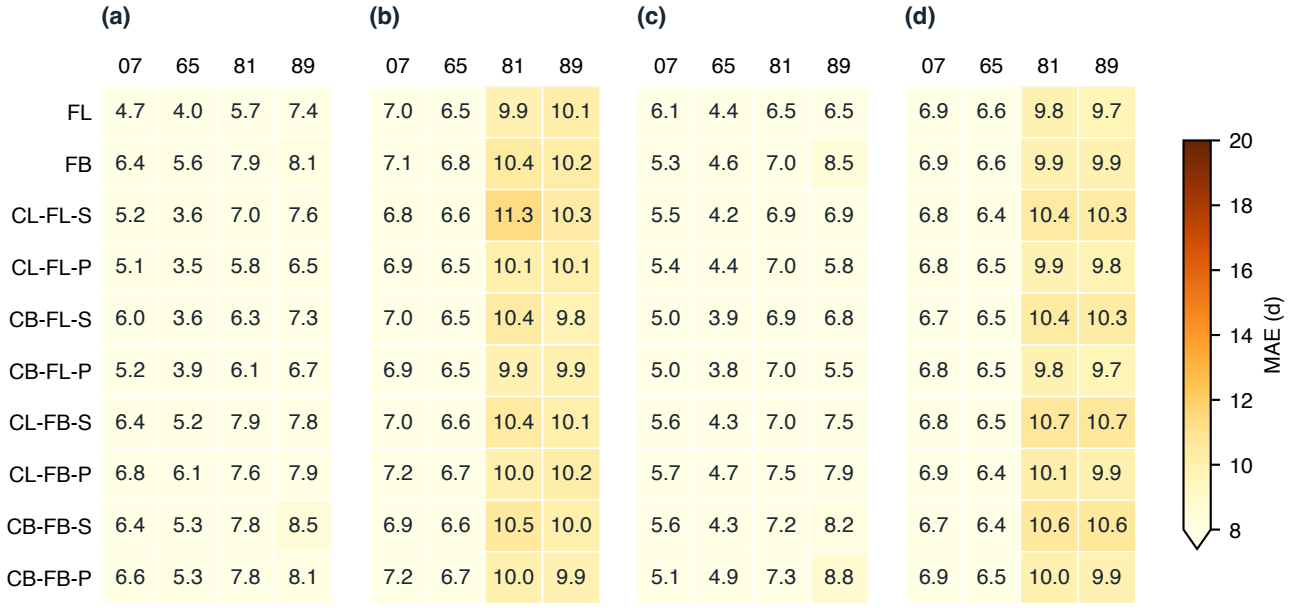
**(a)**

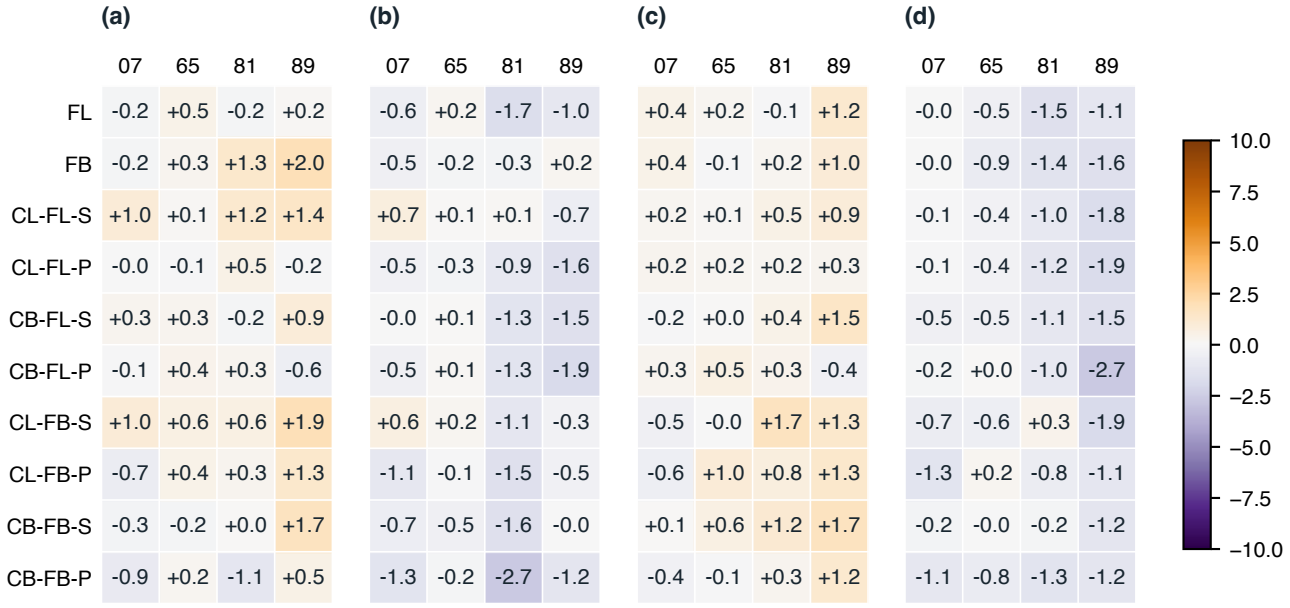
| Country | FL | FB | CL-FL-S | CL-FL-P | CB-FL-S | CB-FL-P | CL-FB-S | CL-FB-P | CB-FB-S | CB-FB-P |
|---|---|---|---|---|---|---|---|---|---|---|
| BA | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| CH | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| CZ | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| DE | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| ES | -2.3 | -2.2 | +0.5 | -0.3 | -0.3 | -0.4 | +2.5 | -0.6 | +0.5 | -2.0 |
| FR | +0.7 | +0.0 | +0.6 | +0.9 | +0.3 | +0.8 | +1.1 | +2.6 | +0.9 | +1.5 |
| HR | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| LU | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| ME | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| SE | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| SK | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |

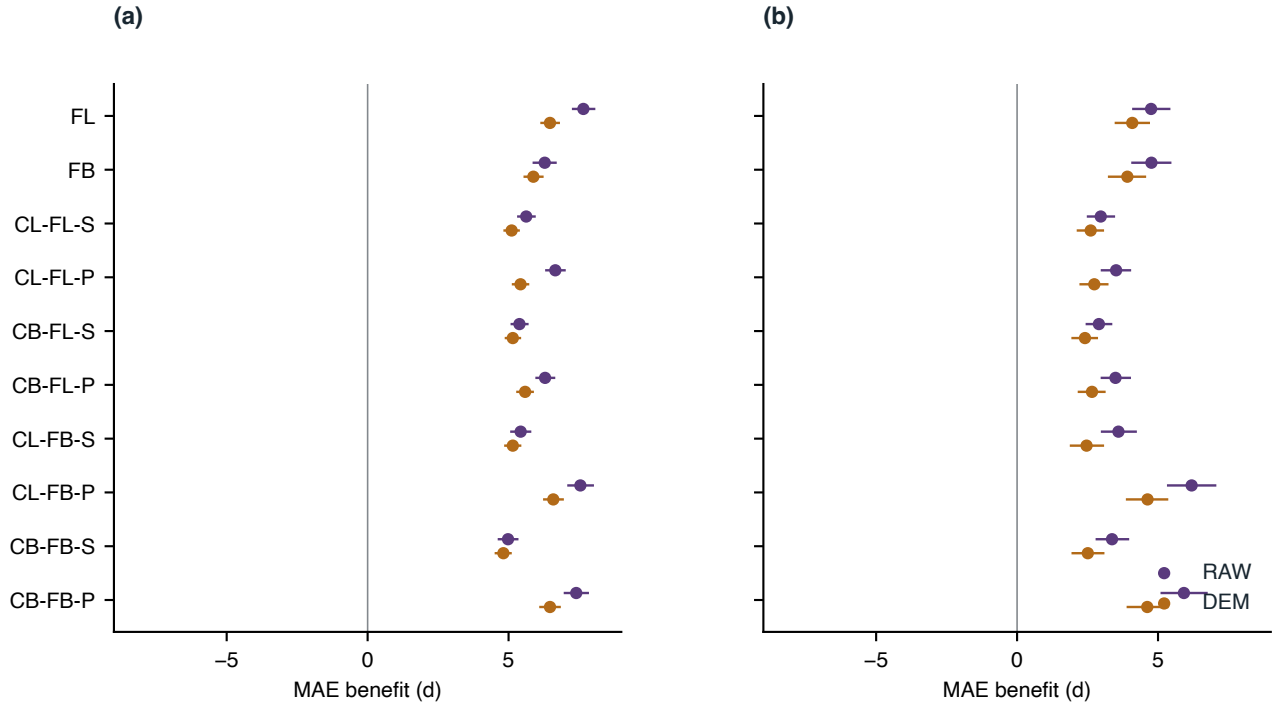
**(b)**

| Country | FL | FB | CL-FL-S | CL-FL-P | CB-FL-S | CB-FL-P | CL-FB-S | CL-FB-P | CB-FB-S | CB-FB-P |
|---|---|---|---|---|---|---|---|---|---|---|
| BA | +3.8 | +5.3 | +2.7 | +3.3 | +3.3 | +3.6 | +3.7 | +3.6 | +3.5 | +4.1 |
| CH | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| CZ | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| DE | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| ES | +1.3 | +1.5 | +0.6 | +1.2 | +0.5 | +1.4 | +2.4 | +5.0 | +2.2 | +3.2 |
| FR | +0.8 | +0.1 | +0.9 | +1.0 | +0.4 | +0.9 | +0.8 | +2.7 | +0.5 | +1.6 |
| HR | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| LU | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| ME | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| SE | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |
| SK | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a | n/a |




(a)
(b)
(c)
(d)
07 65 81 89
FL 4.7 4.0 5.7 7.4 7.0 6.5 9.9 10.1 6.1 4.4 6.5 6.5 6.9 6.6 9.8 9.7
FB 6.4 5.6 7.9 8.1 7.1 6.8 10.4 10.2 5.3 4.6 7.0 8.5 6.9 6.6 9.9 9.9
CL-FL-S 5.2 3.6 7.0 7.6 6.8 6.6 11.3 10.3 5.5 4.2 6.9 6.9 6.8 6.4 10.4 10.3
CL-FL-P 5.1 3.5 5.8 6.5 6.9 6.5 10.1 10.1 5.4 4.4 7.0 5.8 6.8 6.5 9.9 9.8
CB-FL-S 6.0 3.6 6.3 7.3 7.0 6.5 10.4 9.8 5.0 3.9 6.9 6.8 6.7 6.5 10.4 10.3
CB-FL-P 5.2 3.9 6.1 6.7 6.9 6.5 9.9 9.9 5.0 3.8 7.0 5.5 6.8 6.5 9.8 9.7
CL-FB-S 6.4 5.2 7.9 7.8 7.0 6.6 10.4 10.1 5.6 4.3 7.0 7.5 6.8 6.5 10.7 10.7
CL-FB-P 6.8 6.1 7.6 7.9 7.2 6.7 10.0 10.2 5.7 4.7 7.5 7.9 6.9 6.4 10.1 9.9
CB-FB-S 6.4 5.3 7.8 8.5 6.9 6.6 10.5 10.0 5.6 4.3 7.2 8.2 6.7 6.4 10.6 10.6
CB-FB-P 6.6 5.3 7.8 8.1 7.2 6.7 10.0 9.9 5.1 4.9 7.3 8.8 6.9 6.5 10.0 9.9
20
18
16
14
12
10
8
MAE (d)


(a)
(b)
(c)
(d)
07 65 81 89
FL -0.2 +0.5 -0.2 +0.2 -0.6 +0.2 -1.7 -1.0 +0.4 +0.2 -0.1 +1.2 -0.0 -0.5 -1.5 -1.1
FB -0.2 +0.3 +1.3 +2.0 -0.5 -0.2 -0.3 +0.2 +0.4 -0.1 +0.2 +1.0 -0.0 -0.9 -1.4 -1.6
CL-FL-S +1.0 +0.1 +1.2 +1.4 +0.7 +0.1 +0.1 -0.7 +0.2 +0.1 +0.5 +0.9 -0.1 -0.4 -1.0 -1.8
CL-FL-P -0.0 -0.1 +0.5 -0.2 -0.5 -0.3 -0.9 -1.6 +0.2 +0.2 +0.2 +0.3 -0.1 -0.4 -1.2 -1.9
CB-FL-S +0.3 +0.3 -0.2 +0.9 -0.0 +0.1 -1.3 -1.5 -0.2 +0.0 +0.4 +1.5 -0.5 -0.5 -1.1 -1.5
CB-FL-P -0.1 +0.4 +0.3 -0.6 -0.5 +0.1 -1.3 -1.9 +0.3 +0.5 +0.3 -0.4 -0.2 +0.0 -1.0 -2.7
CL-FB-S +1.0 +0.6 +0.6 +1.9 +0.6 +0.2 -1.1 -0.3 -0.5 -0.0 +1.7 +1.3 -0.7 -0.6 +0.3 -1.9
CL-FB-P -0.7 +0.4 +0.3 +1.3 -1.1 -0.1 -1.5 -0.5 -0.6 +1.0 +0.8 +1.3 -1.3 +0.2 -0.8 -1.1
CB-FB-S -0.3 -0.2 +0.0 +1.7 -0.7 -0.5 -1.6 -0.0 +0.1 +0.6 +1.2 +1.7 -0.2 -0.0 -0.2 -1.2
CB-FB-P -0.9 +0.2 -1.1 +0.5 -1.3 -0.2 -2.7 -1.2 -0.4 -0.1 +0.3 +1.2 -1.1 -0.8 -1.3 -1.2
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
Bias (d)


(a)
(b)
FL
FB
CL-FL-S
CL-FL-P
CB-FL-S
CB-FL-P
CL-FB-S
CL-FB-P
CB-FB-S
CB-FB-P
−5
0
5
MAE benefit (d)
RAW
DEM